Supplementary Material

Certified inverse control of separatrix transport in a parametrically driven Kerr oscillator

A. M. Tishin

Supplementary material for Nonlinear Dynamics. A. M. Tishin, Lomonosov Moscow State University and Moscow Institute of Physics and Technology. Correspondence: tishin@amtc.org.

Appendix A derives the controlled Hamiltonian. Appendix B gives the sectorial coefficient calculation. Appendix C documents the numerical transport tests. Appendix D gives a uniform upper bound and states the matching requirements. Appendix E gives the interval proofs for the contour, persistence and phase control, with reproduction commands. Equation and bibliography numbers are local to this supplement. E.12 gives the positive-phase cusp and the explicit equal-share inverse example.

Appendix A Controlled Hamiltonian

A.1 Monochromatic channels

In the laboratory frame let $n = (q^2 + p^2)/2$. The controlled Hamiltonian is

$$\begin{aligned} H_{\rho,\phi} &= (1 + \delta + \epsilon\cos 2t)n + U/2\, n^2 \\ &+ \epsilon/4\,[(q^2 - p^2)\cos 2t - 2qp\sin 2t] \\ &+ \rho\epsilon/4\,[(q^2 - p^2)\cos(2t + \phi) + 2qp\sin(2t + \phi)]. \end{aligned} \qquad (A1)$$

The parameters ρ and ϕ specify the counter-rotating channel. With $q = \sqrt{2J}\cos\theta$, $p = -\sqrt{2J}\sin\theta$ and $\psi = \theta - t$, the resonant contribution is $(\epsilon J/2)\cos 2\psi$, and the counter-rotating contribution is $(\rho\epsilon J/2)\cos(2\psi + 4t + \phi)$. The bracket of the latter conjugate pair gives the static correction $-\rho^2\epsilon^2 J/16$ under second-order averaging. The trace component has a zero self-bracket, and its temporal index differs from that of the counter-rotating component.

Write $V_{\pm 4} = (\rho\epsilon J/4)e^{\pm i(2\psi+\phi)}$ for this conjugate pair. In the convention $\{\psi, J\} = 1$, its second-order contribution is $i\{V_4, V_{-4}\}/4 = -\rho^2\epsilon^2 J/16$.

The two traceless quadratic controls have the form $I(t)(q^2 - p^2)/2 - Q(t)qp$, with

$$I(t) = \epsilon/2\,[\cos 2t + \rho\cos(2t+\phi)],$$
$$Q(t) = \epsilon/2\,[\sin 2t - \rho\sin(2t+\phi)]. \qquad (A2)$$

With s the fast phase on the extended system, put $w = \epsilon e^{2i\psi}/4$, $E = e^{4is+i\phi}$ and $N = J$. The exact rotating Hamiltonian is

$$K = U/2\,N^2 + \delta N + \epsilon N\cos 2s + N\big(w + \epsilon^2/(16w)\big) + \rho N\big(wE + \epsilon^2/(16wE)\big). \qquad (A3)$$

The bracket $\{w, N\} = 2iw$ gives, with $D = \partial_\tau + \partial_s$,

$$DN = -2iNw(1+\rho E) + R_N,$$
$$Dw = 2iw[UN + w(1+\rho E) + \epsilon\cos 2s] + R_w,$$
$$R_N = i\epsilon^2 N(1+\rho/E)/(8w),$$
$$R_w = 2i\delta w + i\epsilon^2(1+\rho/E)/8. \qquad (A4)$$

For $\delta = d\epsilon^2$, $|N| \le M_0$, $m \le |w| \le M_0$ and $|\mathrm{Im} s| \le a$, put $B_0 = 1 + |\rho|e^{4a}$. Then $|R_N| \le \epsilon^2 M_0 B_0/(8m)$ and $|R_w| \le \epsilon^2(2|d|M_0 + B_0/8)$. For two trajectories in this compact domain on a characteristic segment of length T, a Lipschitz constant L for the reduced field and residual bound $C\epsilon^2$ give

$$\| \Delta Z(T) \| \le e^{LT} \| \Delta Z(0) \| + C\epsilon^2(e^{LT} - 1)/L. \qquad (A5)$$

For $L = 0$ the quotient is T. The estimate uses a fixed lower bound on $|w|$ and a prescribed initial discrepancy. In an expanding matching region w tends to zero, and the discrepancy of the true invariant manifolds must be controlled on the splitting scale.

A.2 Range of the control comparison

Additional temporal harmonics give an integrable benchmark. Set $\chi(t) = t + (\epsilon/2)\sin 2t$, retain $(1 + \delta + \epsilon\cos 2t)J + UJ^2/2$ and replace the angular pump by $(\epsilon J/2)\cos\big(2\theta - 2\chi(t)\big)$. The change $\psi = \theta - \chi(t)$ subtracts $\dot{\chi}J$ and gives the autonomous Hamiltonian $\delta J + UJ^2/2 + (\epsilon J/2)\cos 2\psi$ with coincident separatrices. The comparison in Section 3 of the main article therefore uses the fixed monochromatic budget.

For $U \neq 0$ and $\epsilon > 0$, no time-dependent real linear symplectic change makes Eq. (A1) autonomous. If $(q,p)^T = S(t)Z$, the quartic is $U(Z^T S^T S Z)^2/8$ and the extra generator is quadratic. Autonomy requires $S^T S$ constant. A fixed symplectic change reduces the remaining transformation to a rotation. For $\rho \neq 0$ the squared modulus of the traceless coefficient is

proportional to $1+\rho^2+2\rho\cos(4t+\phi)$ and a rotation preserves it. For $\rho=0$ freezing the traceless term requires uniform rotation at the resonant frequency, leaving the time-dependent trace term. This argument concerns linear canonical changes.

Appendix B Inner coefficients and sectorial projection

B.1 Leading coefficient by residues

Set $\phi=0$, $z=e^{2is}$ and $\delta=d\epsilon^2$. The leading zero harmonic is $X_0=(N_0,w_0)=\big(i/(U\tau),-i/(2\tau)\big)$. The harmonic operator is $\partial_\tau+2ik-M/\tau$, with

$$M=\begin{pmatrix}-1 & 2/U\\ U & 0\end{pmatrix},\qquad v_+=(1,U)^T,\quad v_-=(1,-U/2)^T. \tag{B1}$$

The eigenvalues are 1 and -2, and the zero-index fundamental matrix has columns τv_+ and $\tau^{-2}v_-$. At first order the first harmonic receives the direct trace contribution and the interaction of the index $+2$ counter-rotating response with the index -1 trace response. At $U=-1$ the direct projected source is $-1/(3\tau^2)$, giving $C_{\mathrm{dir}}=4\pi/3$ on the oriented jump contour.

Let $X=(X_N,X_w)$ be the index $+2$ response per unit ρ and $Y=(Y_N,Y_w)$ the index -1 trace response. Their ray representations for $U=-1$ are

$$\begin{gathered}A=\int_0^\infty e^{-4y}\,(\tau+iy)^{-3}\,dy,\qquad B=\int_0^\infty e^{-2z'}\,(\tau-iz')^{-2}\,dz',\\ X_N=2/3\,\tau A+1/12\tau^2\,,\quad X_w=-2/3\,\tau A+1/24\tau^2\,,\\ Y_N=-i/3\,\tau B+i/6\tau+1/12\tau^2\,,\quad Y_w=i/3\,\tau B+i/12\tau+1/24\tau^2\,.\end{gathered} \tag{B2}$$

The integrals are continued laterally. Substitution into the projected mixed source yields

$$\begin{gathered}G=A\big(-8\tau B/9+4i/9-1/(9\tau)+i/(18\tau^2)\big)\\ +B\big(-5i/(9\tau)+1/(36\tau^2)\big)-i/(4\tau^3)-1/(9\tau^4).\end{gathered} \tag{B3}$$

The integral of $e^{2i\tau}G$ reduces to residues at $\tau=0,-iy,+iz'$. On a finite contour at $\mathrm{Im}\tau=-c$, the pole $-iy$ contributes for $y<c$. The weighted tail vanishes as $c\to\infty$. After division by $2\pi i$ and then by i, the four contributions are

$$\begin{gathered}I_A=-10/27,\qquad I_B=17/18-(8/9)\ln2,\\ I_{AB}=-4/9+(8/9)\ln2,\qquad I_0=35/54.\end{gathered} \tag{B4}$$

For example $y=uv$, $z'=u(1-v)$ reduces the mixed residue to

$$I_{AB} = 8/9 \int_0^\infty [e^{-2u}(1 - 1/2u^2) + e^{-4u}(1/u + 1/2u^2)]\, du \qquad (B5)$$
$$= 8/9\,(\ln 2 - 1/2).$$

The combined integrand is integrable at zero. The sum is $7/9$, so $C_{\mathrm{cross}} = -14\pi/9$ and

$$C_1(\rho) = 4\pi/3\,(1 - 7/6\,\rho). \qquad (B6)$$

B.2 Formal matching data and the third-order hierarchy

The grading for $\epsilon^n \rho^m z^k$ is generated by ρz^2, ϵz, ϵz^{-1}, ϵ^2 and $\epsilon^2 \rho z^{-2}$. With their multiplicities A', B', C', D', E', one has $n = B' + C' + 2D' + 2E'$, $k = 2A' + B' - C' - 2E'$ and $m = A' + E'$. Thus $n - k$ is even, $k \geq -n$, and $m \leq n$ for $k = 1$. The formal first harmonic has odd orders and at most cubic dependence on ρ at third order under graded matching data.

Write Z_0 for the $(n, k) = (2,0)$ block. Its source in τ has coefficients through τ^{-3}

$$b_0 = (-i/(4U),\, i/8)^T, \quad b_{-1} = (0,\, d - \rho^2/16)^T,$$
$$b_{-2} = (-i\rho(\rho + 4)/(32U),\, i\rho(3\rho - 4)/64)^T, \qquad (B7)$$
$$b_{-3} = (\rho(20 - 9\rho)/(128U),\, \rho(3\rho + 20)/128)^T.$$

The growing part is $\tau b_0/3 + \kappa\tau v_+$. The leading outer resonant separatrix near the upper pole re-expands as

$$N_{\mathrm{out}} = \frac{i\epsilon}{U \sinh(\epsilon\tau)}, \qquad w_{\mathrm{out}} = -\frac{i\epsilon}{4} \coth(\epsilon\tau/2). \qquad (B8)$$

The coefficients of $\epsilon^2\tau$ are $-i/(6U)$ and $-i/24$, selecting $\kappa = -i/(12U)$ in this decomposition. A higher outer term of order ϵ^{m+1} with local power p contributes $\epsilon^{m+1+p}\tau^p$. Hence only $m = 0$, $p = 1$ contributes $\epsilon^2\tau$. This selects the growing datum in formal matching.

The projection $P_- = (I - M)/3$ gives $P_- b_{-3} = (-\rho^2/(16U), \rho^2/32)^T$, the coefficient of $\log(\tau)/\tau^2$ in Z_0. It remains nonzero at $\rho = 6/7$. For the growing variational mode with carrier $e^{-2i\tau} e^{2is}$, the corresponding constant source at second order is $\rho(4 - \rho)v_-/16$. Its $P_+ = (M + 2I)/3$ projection vanishes, eliminating the resonant $\epsilon^2\tau \log\tau$ in this mode.

The third-order calculation uses the following ordered blocks

$$(0,k): k = 2,4,\ (1,k): k = -1,1,3,$$
$$(2,k): k = -2,0,2,\ (3,k): k = -3,-1,1. \qquad (B9)$$

Each source is obtained from Eq. (A4) by extracting $\epsilon^n z^k$ and moving the linear term $MY_{n,k}/\tau$ to the left. The rational term N/w is expanded about N_0, w_0 through first order in ϵ for the third-order source. These 11 blocks close by $k \geq -n$. The archived hierarchy gives the expanded vectors, with the lower blocks cross-checked against the second-order calculation.

B.3 Sectorial solutions and jump extraction

With $x = i\tau$ the operator is $L_k = \partial_x + 2k - M/x$. Its propagator is

$$K_k(x, y) = e^{-2k(x-y)}[(x/y)P_+ + (y/x)^2 P_-]. \quad (B10)$$

In the upper half-plane the solution for $k > 0$ is $\int_0^\infty K_k\,(x, x-t)G_k(x-t)\,dt$, where G_k is the transformed source. For $k < 0$ use $-\int_0^\infty K_k\,(x, x+t)G_k(x+t)\,dt$. The exponential kernels and the polynomial or logarithmic growth of the preceding blocks on fixed horizontal strips give convergence and uniqueness in the prescribed growth class. For $k = 0$, use $D_0 = d - \rho^2/16$ and the right-end reference

$$\begin{aligned} Z_{\mathrm{ref}}(x) = x(-1/(6U), -1/24)^T + (-D_0/U, -D_0/2)^T \\ +(\rho^2/(16U), -\rho(\rho+4)/64)^T/x \\ +(\rho^2/(16U), -\rho^2/32)^T \log(x)/x^2 \\ +(\rho(20-\rho)/(384U), \rho(20-\rho)/384)^T/x^2 + \gamma v_-/x^2. \end{aligned} \quad (B11)$$

The zero mode is $Z_{\mathrm{ref}} - \int_0^\infty K_0\,(x, x+t)(G_0 - L_0 Z_{\mathrm{ref}})(x+t)\,dt$. The subtracted source is $O(x^{-4})$ at the right end and the kernel grows at most quadratically. The correction is therefore $O(x^{-3})$. We use $\gamma = 0$ unless specified. The lower branch is the coefficientwise complex conjugate on the positive real axis. The linear equations continue both solutions to $x > 0$ away from the pole.

Let R and L denote the upper and lower solutions and let $\Delta = L - R$. Define $S_R = \det(\Delta, \partial_x R)/w_R$ and S_L with L in place of R. The vector field in x preserves density $1/w$, since the trace of its derivative equals the velocity of w divided by w. The x derivative of either solution solves its variational equation. Thus

$$(\partial_x - i\,\partial_s)S_R = \det(G(L) - G(R) - DG(R)\Delta, \partial_x R)/w_R. \quad (B12)$$

The quadratic terms in Δ cancel in the derivative of $(S_R + S_L)/2$. The remainder contains at least three factors of Δ or its derivatives. The minimum-index blocks $(1,-1), (2,-2), (3,-3)$ have

common solutions by the right-end integral construction. Every nonzero difference has weight $n + k \geq 2$. The cubic remainder has weight at least six, so it vanishes in $(1,1)$ and $(3,1)$. Consequently

$$C_n = \frac{e^{2x}}{3i}[(S_R + S_L)/2]_{n,1}, \qquad n = 1,3, \qquad (B13)$$

is constant on $x > 0$. The factor $3i$ comes from pairing the jump $C(-ix)v_+e^{-2x}$ with $\partial_x X_0 = v_-/(Ux^2)$ and dividing by $w_0 = 1/(2x)$. Numerically Eq. (B13) equals the real part of the expression for S_R.

B.4 Pole normalization and detuning

Changing γ translates x to $x - \gamma\epsilon^2$. The exponential carrier gives $C_3 \mapsto C_3 + 2\gamma C_1$, so the value at the leading zero is invariant under this translation.

To compute the detuning response at fixed pole normalization, let X_2, Y_-, Y_+ denote $(0,2), (1,-1), (1,1)$, with X_2 including ρ. Write Z_+ for $(2,2)$ and T_-, T_+ for $(3,-1), (3,1)$. Set

$$q_d = (-1/U, -1/2)^T, \qquad B_d = \begin{pmatrix} 0 & 0 \\ -U/2 & -1 \end{pmatrix}, \qquad (B14)$$

and $Q(X,Y) = \left(0, -U^2 X_N Y_N - U(X_N Y_w + X_w Y_N)\right)^T$. The source derivatives give

$$\begin{aligned} \partial_d Z_0 &= q_d, \qquad \partial_d Z_+ = xB_d X_2, \\ \partial_d T_- &= xB_d Y_-, \qquad \partial_d T_+ = xB_d Y_+ + x^2 Q(X_2, Y_-). \end{aligned} \qquad (B15)$$

The blocks $(2,-2)$ and $(3,-3)$ have zero response. Substitution verifies the identities in the respective equations, and the growth and pole conditions select these solutions. At large x, the jump of X_2 carries e^{-4x} and that of Y_- vanishes. The leading jump of Y_+ has a growing part proportional to xv_+e^{-2x} and a tangent part proportional to $x^{-2}v_-e^{-2x}$. The identity $\det(B_d v_+, v_-)/\det(v_+, v_-) = -1$ cancels the growing contribution against $-q_{d,w}/w_0 = x$ from the denominator. The tangent contribution to $\partial_d C_3$ tends to zero. By Eq. (B13), $\partial_d C_3 = 0$ within this normalization. Detuning can change the outer pole location and the full-system prefactor.

At $U = -1$, $\rho = 6/7$, sectorial integration gives $C_3 = -0.214283275340686$ .... Refining the contour changes its real value by about 9.6×10^{-30}. The absolute first-order calibration errors are about 3.1×10^{-27} at $\rho = 0$, 5.9×10^{-30} at $\rho = 1$, and 1.2×10^{-57} at $\rho = 6/7$ in the

refined run. The test $\gamma = 1/4$ preserves the compensated value and gives the predicted shift $C_1/2$ at $\rho = 0$. Runs with $d = -13/784$ and $d = 1$ check the detuning identity numerically. These differences quantify numerical convergence. An interval enclosure for C_3 remains open.

Appendix C Numerical checks of compensation and transport

C.1 Local harmonic and finite-depth settings

The local diagnostic is $h_F us$ near the first return, using Floquet coordinates u, s and exponent h_F. Here W_1 is the amplitude of its first sampled harmonic. The MPFR Taylor integrations for $\rho = 6/7$ use about 40 decimal digits plus guard bits, order 36, 12 steps per pump period, eight phases and a tangent launch of amplitude 10^{-10}. Selected runs refine the precision, time step and launch amplitude. The reference $W_{\mathrm{ref}} = 4\pi\epsilon \sinh(\alpha/h)/\sinh(\pi/h)$ uses $\alpha = \arccos(2\delta_{\mathrm{eff}}/\epsilon)$, $\delta_{\mathrm{eff}} = \rho^2\epsilon^2/16 - \delta$ and $h^2 = (\epsilon/2)^2 - \delta_{\mathrm{eff}}^2$.

Table C1. Residual at the fixed leading setting with $\phi = \delta = 0$.

ϵ	W_1 **at** $6/7$	$W_1/(\epsilon^2 W_{\mathrm{ref}})$
0.06	2.04724×10^{-27}	0.049800
0.08	2.32485×10^{-21}	0.049301
0.10	1.15694×10^{-17}	0.048777
0.12	3.71356×10^{-15}	0.048229
0.16	5.97245×10^{-12}	0.047058
0.20	5.75790×10^{-10}	0.045791

Fits with linear and quadratic depth corrections give residual powers about 2.022 and 1.998. These model-dependent estimates support two extra powers. The formal inner prediction for the limiting ratio is $|C_3|/(4\pi/3) \simeq 0.05115636$, contingent on matching this normalization to the true harmonic. At $\epsilon = 0.20$ the static-matched choice of δ gives $W_1 = 5.35912 \times 10^{-10}$ at $6/7$.

Table C2. Roots of a signed first-harmonic projection with $\phi = \delta = 0$.

ϵ	**Numerical setting** ρ_0	$(\rho_0 - 6/7)/\epsilon^2$
0.04	0.857072765073	−0.043807543666
0.08	0.856863269903	−0.043685506152
0.12	0.856516701861	−0.043483005714
0.20	0.855429150291	−0.042842671299

At $\epsilon = 0.04$ we use 85 decimal digits plus guard bits, order 56, 16 steps per period and launch amplitude 10^{-16}. One matched-phase sample changes relatively by 1.0×10^{-26} on refinement. Quadratic interpolation in ϵ^2 of the three smallest depths gives intercept -0.043848312379. Cubic interpolation of all four gives -0.043848311730. The extrapolations check the predicted -0.043848311776 within those models. The roots concern the sampled diagnostic.

The checks at Eq. (3) of the main article use 48 decimal digits plus guard bits, order 44, 16 steps per period and eight phases. They evaluate the predicted value of ρ directly. At the separately tuned setting for $\epsilon = 0.20$, a 16-phase calculation with 50 decimal digits gives residual $W_1 \simeq 1.09 \times 10^{-19}$ and $W_2 \simeq 1.16431 \times 10^{-12}$. The tiny first residual is a property of the local diagnostic.

Table C3. First-harmonic amplitude from the full flow at $\phi = \delta = 0$.

ϵ	ρ_{pred}	W_1 **at** $6/7$	W_1 **at** ρ_{pred}	**Reduction**
0.08	0.856862227947	2.32485×10^{-21}	8.66515×10^{-24}	268.3
0.20	0.855388924672	5.75790×10^{-10}	1.35238×10^{-11}	42.6

C.2 Geometry and channel errors

The reversal identity $H_{-\phi}(q, -p, -t) = H_{\phi}(q, p, t)$ gives $W_{\phi}^{s} = RW_{-\phi}^{u}$ at $t = 0$, where $R(q, p) = (q, -p)$. The gap is $D(q) = p_{\mathrm{out},\phi}(q) + p_{\mathrm{return},-\phi}(q)$. Adjacent roots define lobe boundaries and signed areas $\int D(q)\, dq$. The lobe counts use the fundamental interval of $-P$ on the selected branch. At nominal tuning each incoming lobe has area about 5.82154×10^{-13} and there are two per interval. The single-lobe reduction is about 7.89×10^{4} and the total incoming reduction is about 3.95×10^{4}.

Chebyshev representations of degrees 40 and 56 give the reference and tuned areas in Table C4. At degree 56 the discrepancies from direct point checks are below 2×10^{-24} and 9×10^{-24} for the reference and tuned gaps. At the two positive simultaneous-error points we use 50 decimal digits plus guard bits, order 44, 12 steps per period and degree 48. Point discrepancies of the gap are below 2.5×10^{-21}. The unstable manifold is launched along its Floquet tangent. The discrepancies quantify interpolation convergence. These convergence checks apply to the numerical area evaluations in Table C4. Appendix E validates launch and propagation errors for the lobe-count theorem in its specified control rectangle.

The separate 12-phase scan gives first-harmonic sensitivity about 3.362×10^{-7} per unit ρ and 2.960×10^{-7} per radian. The components are nearly orthogonal. Comparing each with W_2 gives individual scales 3.46×10^{-6} and 3.93×10^{-6} for the first harmonic to stay below the nominal second harmonic. Pure phase and amplitude scans and negative-amplitude combinations are included in the numerical data. Direct geometric areas at nonzero phase are supplied for the two displayed simultaneous-error pairs. The local Fourier flux estimate and geometric area agree at those points to about 3×10^{-9} relative.

The direct predicted-area tests at $\epsilon = 0.20$ were repeated with Chebyshev degrees 40 and 56. At degree 56, using 65 decimal digits plus guard bits, Taylor order 50, 16 steps per period and launch amplitude 10^{-12} changed each new area relatively by less than 3×10^{-15} compared with the nominal degree-56 run. The graph construction and direct integration are identical for the added depth and static-detuning controls.

Table C4. Direct geometric transport at $\epsilon = 0.20$ and $\delta = 0$. The two error pairs are relative to the numerically tuned setting. Phase is measured in radians.

Pump setting	$(\Delta\rho, \phi)$	Incoming lobes	Incoming area	Reduction
Original $\rho = 1$	$\phi = 0$	1	4.5953552×10^{-8}	1
Fixed leading $\rho = 6/7$	$\phi = 0$	1	$5.7579023 \times 10^{-10}$	79.81
Predicted $\rho = 0.855388924672$	$\phi = 0$	1	$1.3523773 \times 10^{-11}$	3398
Numerically tuned	(0,0)	2	$1.1643086 \times 10^{-12}$	39469
Fine error point	$(10^{-4}, 10^{-4})$	1	$4.4805771 \times 10^{-11}$	1026
Coarse error point	$(10^{-3}, 10^{-3})$	1	$4.4787765 \times 10^{-10}$	103

The zero-detuning sweep and matched-detuning comparisons in the main article use the same manifold construction. The frozen parameter file, separate numerical settings and results are in the accompanying archive. For depth 0.24 the degree-32 interpolation on the original span failed to invert. We repeated that degree on a narrower span of 1.5 and checked it against the higher-degree calculation on span 2.

C.3 Numerical tests of the two-harmonic prediction

The calibration uses the 12-phase diagnostic at the numerically tuned center $\rho_0 = 0.8554291502909$. The second harmonic $c = 1.16430864 \times 10^{-12}$ sets the nominal incoming area scale of the surrogate. The previously measured slopes set the two approximate axial scales in Section 4 of the main article. The predicted area is one half the integral of the positive part of the calibrated surrogate over $[0,2\pi]$, so the zero-error area is c. For positive amplitude and phase increments, the diagonal varies $a/(2c)$ and $-b/(2c)$ together along the ray through $\left(1/\left(2\sqrt{2}\right), 1/\left(2\sqrt{2}\right)\right)$. The protocol records the magnitude of the cosine coordinate. Phase reflection and zero mean give the same zero count and positive-part area under reversal of b. Fraction one denotes the reduced boundary on each ray. At the test points we split the graph overlap at stationary gaps and solve for crossings on the resulting intervals. We count incoming lobes from the index differences between a crossing and its image under $-P$. The nominal first-harmonic

coefficient vectors from closure_research/data/robustness.json are rotated with the second harmonic to positive sine. With the phase branch specified in Section 4.2, the amplitude increment gives the dominant positive sine slope 3.362×10^{-7} and the phase increment gives the dominant negative cosine slope -2.960×10^{-7} per radian. The numerical protocol uses their magnitudes to set the two axis scales.

Table C5. Two-harmonic predictions and full-flow lobe counts at depth 0.20. Area is in units of 10^{-12}.

Cut	Boundary fraction	Predicted count	Direct count	Direct area
Amplitude	0.7	2	2	1.73479
Amplitude	1.3	1	1	3.02714
Phase	0.7	2	2	1.73471
Phase	1.3	1	1	3.02692
Diagonal	0.7	2	2	1.31704
Diagonal	0.97	2	2	1.48552
Diagonal	1.03	1	1	1.54272
Diagonal	1.3	1	1	1.81952
Diagonal	0.998	2	2	1.51047
Diagonal	1.002	1	1	1.51444

The initial tests used degree 36, 48 decimal digits plus guard bits, Taylor order 34 and eight steps per period. Selected tests were repeated at degree 48 with 55 decimal digits plus guard bits, Taylor order 42, 12 steps per period and launch amplitude 10^{-12}. The largest relative area change in these paired refinements was 2.4e-06. These checks quantify numerical convergence on the chosen graph charts.

C.4 Sectorial integration parameters

The main run integrates on $\mathrm{Im} x = 3$ with radius 64, step 0.5, Taylor order 36 and 70 decimal digits plus guard bits. The refinement uses $\mathrm{Im} x = 4$, radius 80, step 0.25, order 44 and 90 decimal digits plus guard bits. Each block is integrated in the direction selected by its kernel and

continued to the positive real axis. Initial data for the zero and relevant negative modes use rational asymptotic coefficients and the logarithmic term in Z_{ref}. Comparison points span $6 \leq x \leq 26$. Configurations, output and symbolic checks accompany the numerical archive.

Appendix D Uniform bound and matching requirements

D.1 Upper bound for the full system

Fix $U = -1$, $\delta = d\epsilon^2$, compact real sets for ρ, ϕ, d and a compact regular arc of a leading resonance loop in scaled rotating coordinates. For each fixed $0 < \sigma < \pi$, constants $C_\sigma, \epsilon_\sigma > 0$ exist uniformly in those parameters and the real section phase. For $0 < \epsilon < \epsilon_\sigma$, the primary stable and unstable branches are graphs along the arc. In physical coordinates their transverse gap satisfies

$$\text{gap} \leq C_\sigma \sqrt{\epsilon}\, e^{-(\pi-\sigma)/\epsilon}. \qquad (D1)$$

The area of a lobe contained in the regular chart, and the sum for disjoint such lobes, is bounded by $C_\sigma \epsilon e^{-(\pi-\sigma)/\epsilon}$. This concerns the primary continuation of the local branches.

Set $u = \epsilon t$, $q = \sqrt{\epsilon}(Q\cos t + P\sin t)$, $p = \sqrt{\epsilon}(-Q\sin t + P\cos t)$ and $I = (Q^2 + P^2)/2$. With fast phase $\vartheta = 2t$ and $h = \epsilon/2$ the exact equation is $d(Q,P)/d\vartheta = hX_{\mathcal{H}}$, with

$$\begin{gathered} \mathcal{H} = H_0 + I\cos\vartheta + d\epsilon I \\ + \rho/4\,[(Q^2 - P^2)\cos(2\vartheta + \phi) + 2QP\sin(2\vartheta + \phi)], \qquad (D2) \\ H_0 = -(Q^2 + P^2)^2/8 + (Q^2 - P^2)/4. \end{gathered}$$

The leading averaged field is independent of the control. The right homoclinic loop is

$$Q_0(u) = \frac{\sqrt{2}\cosh(u/2)}{\cosh u}, \qquad P_0(u) = \frac{\sqrt{2}\sinh(u/2)}{\cosh u}. \qquad (D3)$$

The left loop is its negative. The nearest poles have $\mathrm{Im}u = \pm\pi/2$. For $|\mathrm{Im}u| \leq a < \pi/2$ and $v = \mathrm{Re}u$ the squared norm is at most $2/(\cos^2 a \cosh v)$. The leading complex flow thus exists along both loops and the saddle for the required imaginary times. Analytic dependence extends it to a common smaller neighborhood. The full scaled field is polynomial in space, analytic in the parameters and periodic analytic in ϑ, with uniform bounds on smaller compact domains.

Apply the Hamiltonian continuous averaging theorem (Theorem 1 of [1]) with $a = (\pi - \sigma)/2$. The periodic symplectic change is close to the identity. In slow time its autonomous Hamiltonian is $H_0 + O(\epsilon)$ in a fixed smaller complex neighborhood, and the residual field is bounded by $C_\sigma e^{-a/h}$ in the spatial C^1 norm after a Cauchy estimate. The odd Hamiltonian fields and their averaging generators fix the origin. The autonomous saddle retains eigenvalues bounded away from zero in slow time. Its regular energy-level arcs persist by the implicit function theorem outside a small saddle neighborhood and connect the local branches of the same energy inside it.

The exponential dichotomy at the saddle is uniform in slow time. The local invariant graphs depend Lipschitz continuously on the residual in the Lyapunov–Perron construction. Their discrepancies from the coincident autonomous branches have the residual bound. Finite-time continuation in slow time carries the bound to the regular arc. The inverse averaging change has bounded derivatives. Rotation preserves distances and areas, and the return to physical units multiplies lengths by $\sqrt{\epsilon}$ and areas by ϵ. This proves Eq. (D1) and the area bound.

The constants and threshold depend on σ. At fixed positive σ the loss $e^{\sigma/\epsilon}$ dominates every polynomial gain. The comparison of compensated and uncompensated coefficients requires the exact exponential scale.

D.2 Conditions for transfer of the coefficients

Completing the square gives a pendulum-type form on the blown-up cylinder. With $j = J/\epsilon$ and $U = -1$ it is $-(j - \cos 2\psi/2)^2/2 + \cos^{22}\psi/8$. The physical origin is on the zero-action boundary. For the full rotating Hamiltonian $UJ^2/2 + Jf(\psi, t)$, the canonical momentum $P = J + f/U$ gives the potential $-f^2/(2U) - G_t/U$, where $G_\psi = f$, and the exact scalar equation $\ddot{\psi} = f_t + f f_\psi$. After removing the trace modulation by an angle shift, the counter-rotating derivative term in this mechanical potential is of order ρ/ϵ on the slow Hamiltonian scale. Thus a direct mechanical substitution has an unbounded perturbation at nonzero ρ.

In the direct embedding of [2] the perturbation vanishes at zero action and the required nontrivial resonant potential is absent. At the tip considered here the limiting derivative of $-P$ is the identity, whereas the generic map results [3] and the related preprints by Gaidashev and Gonchenko (arXiv:1702.01651) and Gelfreich and Brännström (arXiv:0806.2403) use a nontrivial Jordan block. The singular and degenerate matching results [4, 5] inform the

construction. A leading estimate with a logarithmically small error or a theorem for a nondegenerate coefficient has insufficient accuracy for the compensated order. These comparisons concern the direct embeddings and their stated hypotheses.

For a true normalized first harmonic $g_\epsilon(\rho)$, a sufficient leading estimate would be a uniform complex bound $|g_\epsilon - C_1| \le \eta_\epsilon$ on $|\rho - 6/7| < r$ with $\eta_\epsilon \to 0$. If g_ϵ is holomorphic and real on the real axis in the chosen normalization, Rouché's theorem gives a unique real zero in the disk of radius $r/2$ at small ϵ. Its displacement is at most $\eta_\epsilon/|C_1'|$ and the derivative error on the smaller disk is at most $2\eta_\epsilon/r$ by Cauchy's estimate. The next-order law additionally requires a uniformly matched C_3 and a residual of $o(\epsilon^2)$ after fixing the nonvanishing prefactor.

These estimates require the true stable and unstable solutions on a growing complex overlap domain, their boundary data in the sectorial normalization and control of their difference on the splitting scale. The formal selection and logarithmic cancellations of Appendix B constrain this calculation. Transfer to a geometrically defined splitting function and higher-harmonic estimates for a lobe-area theorem remain open.

Appendix E. Interval certificate for the full-flow lobe-count transition

E.1 Map, branch pair and control domain

We set $\epsilon = 0.20$, $\delta = 0$ and $U = -1$ in the laboratory Hamiltonian of Section 2. A canonical rotation through the unperturbed angle gives the Hamiltonian on the rotating coordinates (q, p), with $n = (q^2 + p^2)/2$ and $\beta = 4t + \phi$,

$$H_{\mathrm{rot}} = \epsilon n \cos(2t) - \frac{n^2}{2} + \frac{\epsilon}{4}(q^2 - p^2) + \frac{\rho\epsilon}{4}[(q^2 - p^2)\cos\beta + 2qp\sin\beta]. \qquad \text{(E1)}$$

We use its original time-dependent flow. Let P denote the time-π map in laboratory coordinates at $t = 0$. The rotation at $t = \pi$ is $-I$, so the rotating time-π map is $T = (-I) \circ P$. This map preserves the chosen outgoing and returning branches. The laboratory map P exchanges the centrally symmetric branches. Since the laboratory Hamiltonian is even in (q, p), P commutes with $-I$ and $P^2 = T^2$. All counts below refer to the selected pair and one fundamental interval for T.

Write $c = (\rho, \phi)$, $\rho_0 = 0.85542915029090$ and

$$v = \frac{\rho - \rho_0}{1.7 \times 10^{-5}}, \qquad w = \frac{\phi}{1.9 \times 10^{-5}}. \qquad \text{(E2)}$$

The certified control domain is the closed rectangle $|v|, |w| \le 1/2$. The phase is in radians. The exact decimal ρ_0 specifies the centre of this rectangle.

The unstable chart $W_c(s)$ satisfies $T_c\big(W_c(s)\big) = W_c(\lambda_c s)$, where $\lambda_c > 1$ and $W_c'(0) = (1, \nu_c)$. Define $s_0(c) = 10^{-12}/\sqrt{1+\nu_c^2}$ and $c^- = (\rho, -\phi)$. With $\mathcal{R}(q,p) = (q,-p)$, the two arcs are parametrized by

$$z_u(x,c) = T_c^{86} W_c(s_0(c)\lambda_c^{2x}), \qquad z_s(y,c) = \mathcal{R}\, T_{c^-}^{92} W_{c^-}\big(s_0(c^-)\lambda_{c^-}^{2y}\big). \qquad \text{(E3)}$$

The reflection identity $T_c^{-1} = \mathcal{R} T_{c^-} \mathcal{R}$ follows from $H_{\mathrm{rot}}(q,-p,-t;\rho,-\phi) = H_{\mathrm{rot}}(q,p,t;\rho,\phi)$. The outgoing fundamental strip is $x \in [-0.88, -0.38]$. The selected returning graph uses the overlapping launch charts within $y \in [-0.615724, -0.002692]$ in the 92-period convention. We write $p_u(q,c)$ and $p_s(q,c)$ for these graphs at common q, and set $D(q,c) = p_u(q,c) - p_s(q,c)$.

E.2 Validated construction of the graph gap

We enclose the monodromy at the saddle with the variational flow and use its separated eigenvalues to continue λ_c and the eigenbasis analytically in the complex controls. We validate the local unstable chart on $|s| \le 0.002$ by bounding the defect of its degree-17 odd polynomial approximation and applying the contraction of $W \mapsto T_c \circ W(\cdot/\lambda_c)$ to the cubic and higher terms. In the adapted chart norm the contraction factor is below 0.70. The uniform launch check covers a complex bidisk of radius 0.015 in each physical control and bounds the effective launch radius after absorbing 67 iterates into the chart by 1.71621×10^{-3}. The analytic chart and tensor interpolation with ten nodes per control give uniform bounds for the launch polynomials. On the unit control polydisk the code bounds the tensor interpolation remainder by $0.024\eta(2+\eta)$, where $R = 0.015/(1.9 \times 10^{-5})$ and $\eta = [R/(R-1)][5/(R-4)]^{10}$. The bound uses the Cauchy contour at radius R and interpolation nodes of absolute value at most four.

We propagate the launch polynomials by the rotating equations using 24 time cells per period and Taylor order 24. The wide panels retain degree 16 in launch and total degree four in the controls. The sparse propagation also removes monomials with launch degree plus four times total control degree above 18 and includes them in the truncation allowance. For each time cell

we bound the defect of the polynomial trajectory in the differential equation, including omitted coefficients and rounding, and propagate the input error. To see the logarithmic-norm bound, let $z_0(t)$ denote the real central trajectory of the polynomial approximation. The cubic vector field is $F_3(z) = (z^{\mathrm{T}}z)J_0z/2$, with $J_0(q,p) = (-p,q)$. Its Euclidean logarithmic norm at z_0 is $|z_0|^2/2$. The operator-norm change in its derivative at a complex displacement of size at most r is bounded by $3(2|z_0| + r)r/2$. The code combines these bounds with the linear-part enclosure on each time cell. In cell-scaled time the resulting rate bound L is below $1/2$. If e is the input error and d the defect bound, the uniform output error bound is $(e + d)/(1 - L)$. The code checks that the bound stays below the 10^{-7} tube radius used in computing L.

For each pair of endpoint arc enclosures, a complex contraction in the launch coordinate inverts the q-projection. The iteration on the normalized launch disk of radius 0.70 has factor below 0.20 and a strict self-map bound. The residual of the approximate inverse controls its error, which is then propagated through the p-projection. Opposite nonzero signs of the two real q-projection derivatives distinguish the outgoing and returning graphs. The resulting error bound for D on each complex polydisk is less than 3.24×10^{-15}. Cauchy's formula bounds every derivative used below. For a scalar remainder bounded by E on the unit polydisk, the bound for derivative multi-index α on a smaller polydisk with radii r_j is $E \prod_j \alpha_j\,!/\left(1 - r_j\right)^{\alpha_j}$.

Fourteen pairs of complex panels have real q-intervals covering $[0.1965,0.2945]$, with centres spaced by 0.007 and complex q radius 0.0052. We split each real interval into 44 slabs, giving 616 slabs. Overlapping launch intervals and strict monotonicity of each projection identify the exact graphs on adjacent panels. At both strip endpoints, the outgoing q-coordinate lies within the first or last verified panel for every control in the rectangle. The outgoing projection is increasing on the whole launch strip, so those endpoint bounds place every possible contact of the selected pair in the covered intervals.

E.3 Exhaustion, exact seam and cubic contacts

The tangencies solve $F(q,c) = \left(D, D_q\right) = 0$, where $c = (v,w)$ is the normalized control. On each closed q-slab the verifier constructs a fixed numerical preconditioner B and bounds $\| I - B\,\partial_c F(q,c) \|_\infty < 0.01350$ over the slab and the entire normalized control square. The map $c \mapsto c - BF(q,c)$ is a contraction on the square. The residual bound b at the proposed center and a Lipschitz bound $k < 1$ put the closed cube of radius $b/(1 - k)$ strictly inside the square and

make it invariant. Banach's theorem gives existence in the cube. The derivative bound on the convex square gives uniqueness there, including outside the cube. It also implies that $B\,\partial_c F$ is invertible, hence so is $J_c = \partial_c F$. The roots agree on overlapping slabs, giving a smooth parametrization $c(q)$. The outgoing launch coordinate $x\big(q, c(q)\big)$ has strictly positive derivative along it.

At a common intersection, applying T sends (x, y) to $(x + 0.5, y - 0.5)$. Interval checks place the image and preimage of the selected returning graph in the appropriate opposite endpoint charts. Since the exact map preserves contact multiplicity, the two ends of the fundamental interval identify the same control. This identification also applies when counting crossings.

The verifier follows $D_{qq}\big(q, c(q)\big)$. It checks its sign outside four strips and a strict derivative sign with opposite endpoint signs in each strip. This gives exactly four zeros, all away from the seam. At such a zero $c' = 0$ and $D_{qqq} \neq 0$. Differentiating $D = D_q = 0$ at the zero gives $D_c c'' = 0$, $D_{qc} c'' = -D_{qqq}$ and $D_c c''' = 2D_{qqq}$. Consequently $\det(J_c[c'', c'''\,]) = 2D_{qqq}^2 > 0$, which gives an ordinary semicubical cusp in control space. Every other contact on the locus is quadratic.

Table E1 gives outward-rounded control enclosures for the four cusps. The two zero-phase enclosures use the separate time-reversal sections, with two-control enclosures for the unfolding Jacobian and D_{qqq}. The positive-phase enclosure solves $D = D_q = D_{qq} = 0$ in a narrow box. The returning coordinate in these narrow two-control panels uses 91 periods and shifts by -0.5 when expressed in the wide 92-period convention. Reflection gives the negative-phase cusp.

The narrow lower-amplitude, upper-amplitude and positive-phase enclosures lie inside the wide cusp strips in the crossing coordinate. The corresponding strip bounds, rounded outward here, are [0.2259318181, 0.2264090910], [0.2706363636, 0.2711136364] and [0.2472500000, 0.2475681819]. The narrow and wide graphs agree on their overlapping launch intervals after the period conversion, and uniqueness on each strip identifies the narrow root with its wide counterpart.

Table E1. Certified control enclosures of the four cusps.

Cusp	$10^6(\rho - \rho_0)$	$10^6\phi$ (rad)
Lower amplitude	[−6.92638338, −6.92638237]	0
Upper amplitude	[6.92646572, 6.92646673]	0
Positive phase	[−0.00001414, 0.00001414]	[7.86766647, 7.86769475]
Negative phase	[−0.00001414, 0.00001414]	[−7.86769475, −7.86766647]

E.4 Simplicity of the contour

In this subsection let $c = (v, w)$ be the normalized control vector and let $g = D_c$ along the solution. The code verifies $\det(g, dg/dq) > 0$ on the whole contact curve. At a tangency the two branches have the same point and tangent, so their common control-dependent coordinate change contributes equally to the two control derivatives. For the symplectic map T the difference of these derivatives in the new graph coordinates is g times the reciprocal of the derivative of the new q-coordinate along the common tangent. That derivative is positive here because both outgoing q-projections are increasing in x. Thus the normal directions agree at the seam. The sign of the quadratic gap term is preserved there by the same positive multiplier and the square of the inverse tangential derivative.

The verifier compares the control boxes of all relevant pairs of slabs, including each slab with itself. Disjoint boxes exclude equality directly. For every overlapping pair it finds a connecting arc with at most one cusp on which g lies in a common open half-plane. On an arc without a cusp, D_{qq} has a fixed sign and c' is proportional to the quarter-turn of g, with nonzero scalar of fixed sign. The projection onto the quarter-turn of the half-plane normal is strictly monotone. For an arc containing one cusp, let g_* be the normal at that cusp. The derivative of the projection onto g_* is proportional to $D_{qq}\det(g_*, g)$. Strict rotation of g within the half-plane makes the determinant change sign at the cusp, and D_{qq} changes sign at the same point. The projection therefore remains strictly monotone through the cusp. These interval tests establish simplicity. Together with the exact seam and full-square uniqueness, they give a single Jordan curve containing all contacts of the specified pair in the control rectangle.

E.5 Lobe counts and simultaneous control tolerance

The verifier isolates four simple crossings at $(\rho_0, 0)$ and two at $(\rho_0, 9.5 \times 10^{-6})$, with all counted roots strictly within the fundamental interval. The reflection $\phi \mapsto -\phi$ exchanges the manifolds and preserves the certified contact curve. The test verifies that the reflected returning coordinate, after a period shift, lies in the same returning graph. The induced reflection of the Jordan curve is a nonidentity involution with fixed points at the two amplitude cusps. An involution of a circle with a fixed point is either the identity or reverses orientation and has exactly two fixed points. The off-axis phase cusps exclude the identity. The bounded component is connected and reflection invariant, so it meets the axis. The only bounded axis interval between boundary points is the interval between the two cusps. It follows that $(\rho_0, 0)$ is an interior control. The upper-edge anchor is exterior because the contact curve lies strictly within the control rectangle.

We use the incoming convention of the area calculation in Appendix C.2. With the outgoing and returning graphs ordered as in E.1 and with q increasing, the signed area between successive crossings is $\int (p^u - p^s)\, dq$. The positive intervals count as incoming for this orientation, and the negative intervals count as outgoing. The exact map identifies the end pair and preserves orientation on the outgoing branch. Reversing the overall orientation interchanges the labels and leaves the count of each type unchanged.

The number of crossings is locally constant off the contour by transversality and the exact seam identification. The interior and the part of the exterior in the rectangle are connected, so the anchor counts give four and two crossings throughout the two regions. Successive simple crossings alternate the sign of the graph gap, with the end pair identified by T. Thus half the crossings correspond to incoming lobes, giving two inside and one outside per fundamental interval of the selected pair.

All 616 certified control boxes are separated by more than 4×10^{-4} in the normalized maximum norm from the rectangle

$$|\rho - \rho_0| \le 2.35 \times 10^{-6}, \qquad |\phi| \le 2.65 \times 10^{-6} \text{ rad.} \qquad \text{(E4)}$$

This rectangle is connected and contains the two-lobe anchor, so any simultaneous changes within these bounds retain two incoming lobes. This conclusion controls their number at the

stated pump depth and detuning. Appendix E.9 certifies the phase response of the two areas in this regime. The numerical comparisons in Section 3 use separately prescribed settings.

E.6 Arithmetic and reproduction

The flow generators use binary128 for the local, axial and narrow panels and a two-component binary64 expansion for the wide panels. TwoSum normalizes the expansion and a fused multiply-add recovers the residual of the leading product. Each retained density or cubic coefficient sums at most 3825 temporal and spatial pairs, and each linear contribution at most 225. The twofold implementation adds 10^{-26} times the absolute residual majorant, an absolute underflow allowance and an upward relative pad of 10^{-22} for its bounds. It accounts separately for input conversion. The chart compositions bound rounding in the coefficient ℓ^1 norm, using NumPy long double with at least a 64-bit significand. Interval sign and inclusion tests use mpmath.iv. The supplied scripts require enabled assertions and the source build uses round-to-nearest, GNU C++ with binary128 and libquadmath, and hardware fused multiply-add for the twofold path.

Here is the roundoff budget for the wide step in transport_wide.cpp. Interpret each stored pair as the exact sum of its two binary64 components. Use round-to-nearest, gradual underflow, and the fused operation in twofold.hpp. Put $u = 2^{-53}$. Away from underflow, a normalized pair satisfies $|a_{\mathrm{lo}}| \le u|a_{\mathrm{hi}}|$. In addition, TwoSum gives the leading sum and its exact residual. Rounding the low-component sum and the addition of that residual incurs at most $4u^2(|a| + |b|)$. In multiplication, the leading fused residual is exact, the next two fused operations incur at most $7u^2|ab|$ including the omitted low-low term, and TwoSum renormalizes the result. For division by the exact integers used in the Taylor recurrence, the initial binary64 quotient has error of order u and the corrected residual has size at most $3u|a|$. The omitted rounding of the correction and the twofold residual operations give a bound $16u^2|a/b|$. For the final divisor $1 - L > 1/2$, the low component of the divisor in the correction is covered by using $32u^2|a/b|$. These bounds follow by applying the binary64 unit-roundoff bound to each low operation and retaining the twofold leading residual.

Let S be the sum over time coefficients of the largest component coefficient norm, and let A be the corresponding sum of the largest linear row norms. The rounded Taylor coefficients are the exact polynomial data for the defect estimate. In each retained density coefficient at most 3825

terms are accumulated. The cubic accumulation adds at most 3825 terms and the linear accumulation at most 225. The shorter operations along each contribution, including normalization by the time index and the endpoint sum of at most 25 coefficients, fit within 100 additional stages. With $32u^2$ per stage, the resulting absolute coefficient-error factor is bounded by $(1 + 32u^2)^{7975} - 1 < 3.2 \times 10^{-27}$. The endpoint, linear and cubic absolute sums are bounded by S, AS and S^3, respectively, since the time step is less than one and the density has a factor of one half. The guard $10^{-26}(S + AS + S^3)$ therefore covers each component, including the lower time orders of the retained defect. The factor two in the defect also covers conversion from component bounds to the Euclidean norm.

The checked rate bound and the step size $h = \pi/24$ give $S < 10$ in the wide run. Indeed, the bound on the central trajectory and the bound on the sum of other coefficient norms each follow from their nonnegative quadratic terms in the rate. The explicit linear harmonics give $A < 10$. With these bounds, the absolute errors from gradual underflow in the signed operations, including propagation through products and sums within one cell, are below the allowance 10^{-280}. A squared-norm underflow may lose an amount of order $\sqrt{2^{-1022}}$ after the square root. It enters the rate bound, where the positive tube radius and the positive linear bound leave room within the relative pad. The nonnegative norm and tail sums use fewer than 10^6 stages in any contribution, giving a relative factor below 4.0×10^{-25} before the pad 10^{-22}. The input conversions are included separately. This error budget concerns the propagation of the arc panels. The graph inversion uses the distinct composition and residual bounds of E.2.

Run sh full_contour/reproduce.sh from the certified_contour directory of the code archive to replay the interval proof using the stored flow panels. The replay regenerates the axial and narrow contact checks, the 14 paired-panel certificates, and full_contour/data/closed-contour-certificate.json. The stitching step checks the hashes of the two input panels for each certificate. To rebuild the proof directly from the differential equations, run FULL=1 sh full_contour/reproduce.sh. In a clean archive copy with the contour panels and certificates removed, this command built the flow integrators, validated 100 local chart samples, regenerated the wide, axial and narrow panels, and reproduced the closed contour and the joint two-lobe tolerance. The earlier diagonal trajectory only proposes wide-panel launch centers. The reproduction record in Online Resource 2 lists the software versions and SHA-256 hashes of the regenerated files.

E.7 Persistence under changes in depth and detuning

Let $\mu = (\epsilon, \delta)$ and $\mu_0 = (0.20,0)$. Keep the control rectangle and the outgoing launch strip of E.1. For nearby μ, include the term δn in (E1) and vary the pump depth in its pump terms. The analytic time map, the simple hyperbolic eigenvalue and the normalized local invariant chart depend smoothly on μ and the controls. The strict inverse and overlap bounds in E.2 on the finite panel cover persist. Hence the continued graphs define a smooth gap on the same compact q-range and control square. The endpoint containment for all controls persists, so this range still covers every potential contact of the selected arcs.

The parameter-dependent implicit function theorem continues the solution of $\left(D, D_q\right) = 0$ near each point of the certified graph. Compactness gives a common neighborhood in μ. Uniqueness on overlaps glues the local solutions, and on the compact complement of a small tubular neighborhood of the base solution graph the norm of $\left(D, D_q\right)$ has a positive minimum at μ_0. Thus no additional contact solution enters the covered range. The strict derivative of the outgoing launch coordinate and its endpoint brackets persist. The exact chart conjugacy still shifts the launch coordinates by $(1/2, -1/2)$ at a common point. The seam inclusions are strict and persist, giving a closed curve on the same fundamental strip.

The four simple zeros of D_{qq} along the solution persist, with no additional zeros by the strict signs away from them. The unfolding Jacobian and the cubic derivative stay nonzero there. The simplicity proof is also open under these perturbations. At each regular point a fixed projection stays strictly monotone on a small arc. Near each moving cusp the strict rotation of the contact normal and the simple sign change of D_{qq} give the monotone projection of E.4. The exact map identification preserves the normal direction at the seam. These neighborhoods give local injectivity on a common neighborhood of the diagonal of the parameter circle. Pairs of points outside that neighborhood form a compact set, and their images have positive minimum separation at μ_0. They remain separated under uniform continuation.

The continued contour is disjoint from the two fixed anchors and from the boundary of the control rectangle. The winding indices of the anchors are unchanged. Their simple crossings continue and no other crossings enter at the anchors, by compactness and exact identification at the ends of the fundamental segment. Transversality transports the counts throughout the two complementary regions. The fixed tolerance rectangle (6) has a positive distance from the base

contour and contains the interior anchor. It stays in the interior by uniform continuity and connectedness. This proves Corollary 1 and gives existence of a neighborhood of μ_0. Appendix E.8 gives an explicit sufficient size.

E.8 Quantitative depth and detuning persistence

Set $s = 10^{-16}$, $\epsilon = 0.20 + s\alpha$ and $\delta = s\beta$, with $|\alpha|, |\beta| \le 1$. Use the same normalized amplitude and phase controls, launch domains and seam as in E.1–E.5. For the returning graph the phase reflection changes the sign of the normalized phase coordinate. Both graphs use the same α, β.

The starting charts use the base depth and detuning and four symmetric shifts of size $h = 10^{-11}$, two along each parameter axis. Interval monodromy bounds on the complex control disks of radius 0.015 give a launch norm below 0.012 at all five settings. On the smaller complex control disks used for the flow panels, the monodromy perturbation bound gives a launch norm below 0.014 for $|\epsilon - 0.20|, |\delta| \le \eta = 8 \times 10^{-6}$. The simple eigenvalue and the invariant-chart contraction give holomorphic dependence on these parameters. The two affine launch coefficients are the symmetric differences multiplied by $s/(2h)$. The normalized tail and difference error are bounded by

$$0.028\left[\frac{t^2(3-2t)}{(1-t)^2} + \frac{2tj^2}{1-j^2}\right], \qquad t = s/\eta, \quad j = h/\eta.$$

The first term sums all bivariate degrees at least two by Cauchy's estimate. The second bounds the odd powers omitted by the central differences along the two axes. The prefactor bounds the sum of the errors in the two launch components. The interval errors of the five samples enter the same affine calculation with their absolute weights.

At zero detuning the linear field is proportional to depth. Its reference polynomial and remainder therefore receive the same affine factor. The detuning adds opposite constants to the two off-diagonal entries of the linear field. The flow solver retains terms through first order in the new parameters. It bounds the residual in the original equations by absolute majorants for the discarded time, launch, control, added-parameter and weighted degrees. In each time cell the checked logarithmic norm and defect propagate the error as in E.2. The twofold arithmetic follows E.6. With launch degree 20 the enlarged accumulation bounds are 9450 terms for each

density or cubic coefficient and 450 for each linear coefficient. With at most 19450 stages and $u = 2^{-53}$, the bound $(1 + 32u^2)^{19450} - 1 < 7.68 \times 10^{-27}$ stays within the 10^{-26} defect guard.

The projection inverse is proposed with its two affine corrections and validated by the substitution residual on the common q disk. In the composition at $z = z_0 + \alpha z_1 + \beta z_2$, the nominal polynomial has a residual at most $\frac{1}{2} \| P_{0,xx} \| (\| z_1 \| + \| z_2 \|)^2$. The affine terms add at most $\sum_{i=1}^{2} \| P_{i,x} \| (\| z_1 \| + \| z_2 \|)$. The exact graph errors include these bounds and the flow and inverse remainders. We add the affine coefficients of the signed gap at common q before taking their norms. Their contribution and the residual form one uniform remainder in the nominal gap. Cauchy estimates apply in q and the two control coordinates at each fixed pair α, β.

The resulting check covers all 14 panels and 616 strips. It verifies the strict contraction, the four simple zeros of the curvature condition, simplicity, the exact seam and the two anchor counts uniformly in the depth and detuning box. The fixed tolerance rectangle (6) has separation greater than 10^{-4} in the normalized maximum norm. Its physical half-widths are the same as in Theorem 1. The anchor argument in E.7 gives the counts throughout the two regions. The explicit radius is a sufficient lower bound for the ideal model. In this verification the additional parameters enter through affine terms and a uniform remainder. The resulting radius records a successful certificate. The maximal radius of contour persistence is undetermined.

From the root of Online Resource 2, run python uniform_contour/uniform/uniform_verify.py certified_contour uniform_contour/arcs uniform-checked and compare uniform-checked/result.json with uniform_contour/certificate/result.json. The archive README gives the commands for generating the launch charts and arc models directly from the equations.

E.9 Certified phase response of incoming areas

Use the band in Theorem 2 at the base depth and detuning. The positive signed gap integral defines the incoming convention of E.5. Fix an amplitude ratio in the band. The count theorem guarantees four simple intersections in the fundamental strip for every phase in the scan. On each of 27 adjacent phase intervals of width 10^{-7} rad, the verifier brackets the four roots by strict opposite signs, uniformly over the amplitude band. Endpoint projections at both launch seams put the brackets strictly inside the fundamental strip. The first intersection is the one with lowest q in that strip.

For integration, start at the first root and finish at its exact image under T. The launch coordinates of the image are obtained by shifting those of the first root by $(+1/2, -1/2)$. Both image launch coordinates are enclosed inside the last outgoing and returning chart domains. For this final panel the q disk has radius 0.0065 in physical q. The inverse contraction and the substitution bound are checked on this larger disk. Sign checks cover the extension to $q = 0.2945$, and a strict gap derivative in q covers the remaining interval to the image. The image lies on both exact graphs and is the only crossing there. Together with the four roots in the original strip, this gives all the successive crossings from the first root to its image.

On each chart interval the gap model is a polynomial in normalized q, amplitude and phase with a uniform error on the unit polydisk. The verifier intersects the possible root-to-root segment with each chart interval, integrates the polynomial between the interval endpoints and adds the interval-length bound times the uniform error. If a chart contribution can be empty, the enclosure also includes zero. Integrating through a join of charts is valid because they enclose the same exact branches there. Each lobe area has the sign in E.5, and the two positive integrals are ordered by their left roots.

The simple roots are differentiable in phase. For consecutive roots $q_a(\phi), q_b(\phi)$ the derivative of their signed integral is

$$\frac{d}{d\phi}\int_{q_a(\phi)}^{q_b(\phi)} D\,(q,\rho,\phi)\,dq = \int_{q_a(\phi)}^{q_b(\phi)} \partial_\phi\, D(q,\rho,\phi)\,dq.$$

The boundary terms vanish at both exact intersections. The formula applies to the image endpoint as well. Differentiating the polynomial in the normalized phase $w = \phi/(1.9 \times 10^{-5})$ and adding the Cauchy remainder bound $e/(1 - |w|)$ encloses the integrand before division by 1.9×10^{-5}. Here e is the chart gap error and $|w|$ is bounded above on the phase interval. The interval derivative of the first incoming area is strictly negative and that of the second strictly positive on every interval. Positivity of both areas then gives strict growth of the second area fraction. The two phase windows in Theorem 2 are checked as complete parameter boxes. The verifier uses monotonicity of $A_i/(A_1 + A_2)$ in each positive area to enclose its share.

From the root of Online Resource 2, run PYTHONPATH=certified_contour/full_contour/code python phase_partition/check_flux.py certified_contour. The checked log, the figure data and its generation commands are included with the code.

E.10 Phase response under joint depth and detuning deviations

Proposition 1 uses $\rho = 0.85542915029090$ and $\phi_0 = 2.65 \times 10^{-6}$ rad with $U = -1$. The real parameter box is $|\epsilon - 0.20| \le 10^{-6}$ and $|\delta| \le 10^{-6}$. Four ordered simple crossings lie on the fundamental segment of the outgoing branch. The first crossing and its image under T are paired by the exact change of the launch coordinates, as in E.9. The four crossing coordinates in increasing order lie in [0.20690495, 0.20700336], [0.2301875, 0.2310625], [0.24725, 0.2476875] and [0.2651875, 0.265625]. The image of the first crossing lies in [0.29661085, 0.29670924].

We validate the two local invariant charts at sparse nodes in depth, detuning and phase and interpolate their coefficients using divided differences. Contour estimates bound the interpolation and Taylor tails. Defect and logarithmic-norm bounds transport the polynomial models through 13 paired panels of the original flow. The returning graph is constructed by reflection and a phase reversal. The unit phase coordinate corresponds to 5×10^{-8} rad. The first phase coefficient of each endpoint polynomial gives its phase derivative at ϕ_0, with a Cauchy bound on the derivative of the error.

If the branch is parameterized by x, projection at common q gives

$$\partial_\phi p \,|_q = p_\phi - p_x \, q_\phi / q_x.$$

Here subscripts denote partial derivatives and the derivative on the left holds q fixed. The inverse projection and reciprocal slope are enclosed on each panel. The signed graph-gap integral is differentiated as in E.9. Its endpoint terms vanish because the gap vanishes at each simple intersection, including the exact image endpoint. The verifier integrates the polynomial models over the common interior of each lobe and adds interval bounds for the two uncertain end segments and the uniform remainder.

Table E2. Lobe areas and phase derivatives at ϕ_0 over the joint depth and detuning box. The positive signed integrals correspond to incoming lobes.

Lobe	Direction	Area (10^{-12})	Phase derivative (10^{-7}/rad)
1	Outgoing	[1.0322, 1.0473]	[1.7792, 2.0794]
2	Incoming	[0.2477, 0.2632]	[−1.0656, −0.8094]
3	Outgoing	[0.2508, 0.2604]	[−1.0209, −0.8788]
4	Incoming	[1.0361, 1.0446]	[1.8877, 2.0522]

Bounds in Table E2 are rounded outward. Adding the derivative intervals for lobes 2 and 4 gives the interval for the total incoming area derivative in Proposition 1. The signs also imply a positive derivative of the share of the second incoming lobe at ϕ_0. This interval is an enclosure over the entire box. Its width incorporates parameter dependence and the successive interval bounds used in the graph and integration calculations. Resolving the actual extrema of the derivative would require tighter parameter-dependent enclosures.

Online Resource 2 contains joint_phase/README.md and the certified models, interval tables and replay code for this calculation. The separate intersection, area and derivative checks can be run from the archived local samples and endpoint models. The same files give the commands for rebuilding the models from the original flow.

E.11 Inverse control and convex transport floor

Theorems 3 and 4 use the original-flow interval panels of *E*.2 and the exact endpoint identification of *E*.9. The two incoming areas A_1 and A_2 are the positive gap integrals with the ordering in *E*.9. In both input boxes the closed-contour certificate gives four simple crossings. The interval verifier brackets the four crossings and checks the exact image of the first on the last panel. That panel is verified on *a* common-*q* disk of radius 0.0085 with normalized launch radius 0.91. The certificate checks monotonicity on the extension to the image and verifies the hashes of all 28 input panels.

For the finite inverse box, the verifier certifies the phase derivative of *F* and the Jacobian determinant on 736 input cells. Their lower bounds are 5.4994×10^{-8} and 5.385×10^{-4} in physical controls. On the lower and upper phase faces *F* is at most 1.238160×10^{-12} and at least 1.285559×10^{-12}, respectively. The lower and upper amplitude faces give *s* at least 0.7828426 and at most 0.7041068, respectively. The monotone level-curve argument in Section 4.4 then proves the finite target result.

For the fold, put $B = A_2 - A_1$, $\beta = 2s - 1$ and $E = B - \beta F$. The verifier uses 81 control cells and encloses *E* and the derivatives needed below. The strictly positive phase derivative of *E* and the face signs give *a* unique phase graph for each target share. For an incoming integral between successive simple roots, its second derivative in two control directions equals the integral of the second gap derivative plus the endpoint correction. The correction is the product of the two first control derivatives of the gap divided by its *q* derivative at the left root, minus the same

expression at the right root. This follows by implicit differentiation of the roots and includes the exact image endpoint.

Along the phase graph at fixed share, the second derivative of F with respect to ρ is enclosed in [0.07674, 0.12270]. The maximum displacement of its minimum from ρ_0 is 8.511×10^{-8}. The bound on curvature therefore gives a rise greater than 3.8047×10^{-15} on both amplitude faces. Across the stated share interval the floor lies in $[1.1642165, 1.1647197] \times 10^{-12}$. Integrating the second derivative also gives the square-root separation from the minimizer on either branch. The four roots remain simple at every control setting used in this argument.

At the constrained minimum write $\nabla F = \lambda \nabla E$ and $\mathrm{H} = \nabla^2 \mathrm{F} - \lambda \nabla^2 \mathrm{E}$. The interval Hessian test gives positive definiteness. If u is the derivative of the minimizing control pair with respect to β, the constraint gives $\nabla E \cdot u = F$. Differentiating stationarity gives the second derivative of the minimum with respect to β as $u^{\mathrm{T}}\mathrm{H}u + 2\lambda^2 \mathrm{F}$. The resulting bound after converting to s proves the stated strict convexity. Positive definiteness of $\nabla^2 \mathrm{F}$ and opposite gradient signs on both pairs of faces give a unique unrestricted minimum in the control box.

For the surrogate gap $\sin(2\theta) + a \sin\theta + b \cos\theta$, the two positive lobes at $a = b = 0$ have unit areas and respective gradients (1,1) and (−1,−1). The total area has zero gradient and identity Hessian, from the moving-endpoint terms. At small imbalance β the minimum total area is $2 + \beta^2/4 + \mathrm{O}(\beta^4)$. The even dependence follows by shifting θ through π, which exchanges the lobes and reverses the first-harmonic coefficients. Thus the ratio of the second derivative of the minimum with respect to s to the minimum itself is one at equal share.

At the unrestricted minimum of the Kerr flow the analogous ratio is 4F divided by the quadratic form of the inverse flux Hessian on ∇B. The small verified box containing the minimum gives [0.99574, 1.00522]. The minimum is interior to both the narrow and extended control boxes in the certificate. Strict convexity identifies the two minimizers.

Online Resource 2 includes inverse_control/README.md, executable verifiers and the interval outputs. The verifiers use the archived enclosures of the original flow, recheck projection and root conditions and integrate the required interval derivatives. The new calculations replay from these enclosures. Equations and transport source for rebuilding the input enclosures are provided with the underlying certificate.

E.12 Phase cusp and an explicit equal-share inverse target

The source for these checks is the original-flow polynomial enclosure of E.2. The wide panels and narrow contact panels in E.3–E.6 cover the same graphs. The replay checks their stored hashes, uses the inverse projection to common q and converts derivatives in normalized coordinates back to physical ρ and ϕ. The gap error on the complex polydisk is propagated by the Cauchy derivative bounds in Appendix E. For second derivatives in controls u and w, each simple endpoint contributes the term from differentiating its root equation, $D_q\, q_u + D_u = 0$. The boundary correction in E.11 is $D_u D_w / D_q$ evaluated at the lower endpoint minus the same expression evaluated at the upper endpoint. The interval implementation is in kerr_control/fold_pilot.py. At the final endpoint the exact launch-coordinate shifts (+0.5, −0.5) place the image of the first root in the extended last panel (E.9). The add_image check in kerr_control/pilot.py encloses that image and tests the sign of D_q across the final part of the segment.

At the cusp. The narrow phase-positive panels enclose the unique solution of $D = D_q = D_{qq} = 0$ in the contact box. The replay checks that D_{qqq} and the control unfolding determinant are positive there. A sign change and a strict negative q derivative isolate the simple root q_1 in [0.2045, 0.209]. The cubic root lies near q = 0.247418. Both lie between the two cuts of the fundamental strip. The same launch-coordinate shift as in E.9 encloses $T(q_1)$ beyond the right cut. An additional simple root would persist on the four-root side where the cubic contact already supplies three roots and q_1 supplies the fourth. An additional multiple root at the same control would contradict the single-contact projection certified in E.3–E.4. Thus the positive gap from the cubic root to $T(q_1)$ defines the surviving incoming lobe. Integrating it with enclosed endpoints gives the interval for A_c in Theorem 5.

Finite phase setting. At ρ_0 and $\phi = 7.70 \times 10^{-6}$ rad, the verifier subdivides the q domain into 1663 checked intervals. It uses strict signs on intervals away from roots and monotonicity with opposite endpoint signs on root brackets. The first root and its image under T close the fundamental segment. For each positive gap interval the verifier integrates across its chart pieces, enclosing the two moving endpoints and adding the uniform chart error. It forms the share directly from the resulting two positive area intervals.

Cusp scaling. Evaluate all derivatives below at the cusp. Put $v_c = D_\phi/D_\rho$, $b = D_{q\rho}\, v_c - D_{q\phi} < 0$, $\kappa = D_{qqq}/6 > 0$ and $a = \sqrt{(-b/\kappa)}$. Take $t = \Delta\phi > 0$, $\rho = \rho_c + v_c t + \eta t^{3/2}$ and $q = q_c + a\sqrt{t}\, y$. Taylor expansion of the analytic graph gap gives

$$t^{-3/2} D = \kappa a^3 (y^3 - y) + D_\rho \eta + O(\sqrt{t}).$$

The remainder and its local y derivatives are uniform on compact y and η intervals. The constant term of order t cancels by the definition of v_c. The coefficient of y combines the mixed q-control derivatives into b. The enclosure gives $D_\rho \in [1.18361, 1.18373] \times 10^{-5}$, $b \in [-7.35132, -7.32416] \times 10^{-4}$ and $D_{qqq} \in [8.58596, 8.58923] \times 10^{-5}$. At $\eta = 0$ the scaled simple roots tend to -1, 0 and 1. Integrating the positive part between the appropriate two adjacent roots, with $dq = a\sqrt{t}\, dy$, gives $A_1 = Ct^2 + O(t^{5/2})$, with $C = \kappa a^4/4 = 3b^2/(2D_{qqq})$. The surviving area tends to A_c by continuity of the graphs and the integration endpoints.

The double roots of the scaled cubic occur at $y = \pm 1/\sqrt{3}$ with $|\eta| = \eta_0 = 2\kappa a^3/(3\sqrt{3}\, D_\rho)$. The equations for a double root have a nonsingular Jacobian in (y, η) there. The implicit function theorem in $\sqrt{t}$ gives two fold curves with $\rho = \rho_c + v_c t \pm \eta_0 t^{3/2} + O(t^2)$. Their half-width is asymptotic to $\eta_0 t^{3/2}$. Since the tangent-ray missed share is asymptotic to $(C/A_c)t^2$, the half-width is asymptotic to $\eta_0 (A_c/C)^{3/4} \ell^{3/4}$. The interval computation encloses η_0 in [170.28, 171.41] and evaluates the coefficient K_{tol} in Theorem 5. These are leading-order bounds as t tends to zero. A finite radius for the tolerance asymptotic would require a quantitative bound on its remainder.

Finite approach with zero gap at the inflection. Use a separate slope parameter σ and solve $D = D_{qq} = 0$ and $D_q = -\sigma$ at a moving q_i for $0 < \sigma \le 8 \times 10^{-14}$. The normalized contraction bound is 0.120280 and the full inclusion bound is 0.668803, enclosing the path in one control box. For $x = q - q_i$, Taylor bounds give $D = -\sigma x + a_x x^3/6$, with a_x between the positive lower and upper bounds for D_{qqq} in the narrow chart. The factor D/x is strictly monotone away from the inflection on each side, by the integral Taylor formula and $D_{qqq} > 0$. Signs at $q_i \pm R$, with $R = 8.5 \times 10^{-5}$, give one exterior root on each side and the inflection is the third root. The positive cubic envelopes give $A_1/\sigma^2 \in [17394.949, 17541.241]$. The monotonicity bridge to the wide panels and their strict sign checks leave only the simple root near q_1 outside the cubic neighborhood on the fundamental strip. The periodic image closes the integration segment and gives $A_2 \in [2.3284044, 2.3288301] \times 10^{-12}$. The resulting missed share satisfies $\ell < 4.822 \times 10^{-11}$. Differentiating the

three path equations gives $d\phi/d\sigma \in [-1366.38, -1359.27]$, yielding $\ell/(\phi_c - \phi)^2 \in [4.0007, 4.0775] \times 10^9$ rad^{-2}.

Equal-share target. The fold calculation in E.11 gives a unique equal-share phase solving B = 0 at each amplitude in the rectangle and a strictly convex F along this graph. At each amplitude face of the two brackets in Section 4.4, we enclose the phase correction by the interval mean-value formula for B. Taylor expansion of F to second order in that correction encloses F at the exact equal-share phase. On the inner amplitude faces the upper bounds lie below the target, and on the outer faces the lower bounds lie above it. The graph has positive phase slope because $B_\rho < 0 < B_\phi$. Thus the phase enclosures at the faces bound the solution in each bracket, and strict convexity ensures that there are exactly two settings in the rectangle.

Reproduction. The README in Online Resource 3 links each assertion to its replay and identifies the input panels and exact-period checks. The base contour can also be regenerated from the ODE using the source build described in Online Resource 2. The cusp and equal-share replays start from those enclosed panels.

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